	do not accept alternative method of finding change in p.e. if not using answer in **(c)(i)**	
4a	thermal energy per unit mass required to produce unit rise of temperature of the substance.	B1

4bi	$p = \frac{1}{3}\frac{Nm}{V}\langle c^2\rangle$ Since pV = NkT, $NkT = \frac{1}{3}Nm\langle c^2\rangle$ $\frac{3kT}{2} = \frac{1}{2}m\langle c^2\rangle$	M1 A1
4bii	Internal energy is the sum of the kinetic energy (due to random motion) of the molecules and the potential energy (due to intermolecular forces) between the molecules. However, for an ideal gas, there is no potential energy due to absence of intermolecular forces between the molecules.	B1 B1
4c	$\frac{1}{2}m\langle c^2\rangle = \frac{3kT}{2}$ $c_{r.m.s} = \sqrt{\frac{3\times1.38\times10^{-23}\times(25+273.15)}{3.34\times10^{-27}}}$ $=1.9\times10^{3}$ m s^{-1}	C1 A1
4di	At constant volume, no work is done by the gas. Thermal energy required to increase the temperature of unit mass of gas by one unit is solely to increase its internal energy. At constant pressure, work is done by the gas as it expands. Thermal energy is required for the increase of its internal energy and the work done by the gas. Hence the specific heat capacity of ideal gas at constant pressure is greater than the specific heat capacity at constant volume.	B1 B1
4dii	Small volume change/decrease and hence work done on ice is negligible/positive, <u>Thermal</u> energy is absorbed to break lattice structure / increase molecular potential energy. Hence, by first law of thermodynamics, internal energy increases.	B1 B1
5a	As current passes through the filament	